

## FLORES AP, Portugal

WMO: 085010

Lat: 39.4572N

Lon: 31.1308W

Elev: 94

StdP: 14.65

Time Zone: -1.00 (AZR)

Period: 96-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 48.8       | 50.3 | 37.9                          | 33.6 | 51.8 | 39.6 | 35.9 | 52.8 | 31.8                  | 58.5 | 29.2 | 60.3 | 10.3                  | 240  | 0.388 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 8.7                    | 80.4            | 72.3 | 78.9 | 71.9 | 77.4 | 71.0 | 73.7                | 78.0 | 72.9 | 77.2 | 72.0 | 76.3 | 7.6                  | 0    |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 72.0                            | 118.9 | 75.9 | 71.4 | 116.3 | 75.7 | 70.2 | 111.7 | 75.1 | 37.1          | 78.0 | 36.4 | 77.5 | 35.7 | 76.4 | 78.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |      |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |      |
| 27.3              | 24.1 | 21.3 | DB                         | 45.5 | 84.0               | 1.3 | 1.4                                                | 44.6 | 85.0       | 43.8 | 85.9       | 43.1 | 86.6       | 42.1 | 87.6 |
|                   |      |      | WB                         | 41.4 | 75.9               | 1.2 | 1.1                                                | 40.5 | 76.6       | 39.8 | 77.2       | 39.1 | 77.8       | 38.3 | 78.6 |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 64.3        | 59.0  | 58.4  | 58.5  | 59.9  | 62.7  | 66.7  | 71.4  | 73.4  | 71.2  | 66.9  | 63.2  | 60.0  |
|                                                                        | DBStd       | 6.19        | 3.46  | 3.32  | 3.16  | 2.96  | 3.19  | 3.22  | 3.07  | 2.73  | 2.92  | 3.58  | 3.65  | 3.75  |
|                                                                        | HDD50       | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 1088        | 187   | 185   | 200   | 154   | 86    | 19    | 1     | 0     | 1     | 21    | 75    | 159   |
|                                                                        | CDD50       | 5230        | 279   | 235   | 265   | 298   | 395   | 501   | 662   | 727   | 637   | 525   | 397   | 309   |
|                                                                        | CDD65       | 843         | 1     | 0     | 0     | 1     | 16    | 70    | 198   | 262   | 188   | 81    | 22    | 4     |
|                                                                        | CDH74       | 1812        | 0     | 0     | 0     | 2     | 9     | 32    | 469   | 928   | 337   | 34    | 1     | 0     |
|                                                                        | CDH80       | 53          | 0     | 0     | 0     | 0     | 2     | 0     | 9     | 38    | 4     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 9.7         | 11.4  | 11.2  | 11.5  | 10.7  | 9.0   | 8.5   | 7.0   | 7.2   | 8.7   | 9.8   | 10.2  | 11.7  |
| (15) Precipitation                                                     | PrecAvg     | 61.6        | 6.8   | 5.9   | 6.0   | 3.9   | 3.9   | 3.7   | 2.6   | 2.8   | 4.7   | 7.1   | 6.1   | 8.1   |
|                                                                        | PrecMax     | 81.4        | 12.7  | 14.0  | 15.0  | 11.9  | 10.5  | 9.5   | 6.2   | 6.3   | 8.9   | 15.9  | 10.2  | 17.3  |
|                                                                        | PrecMin     | 38.3        | 2.2   | 0.8   | 1.9   | 0.1   | 0.9   | 0.4   | 0.4   | 0.5   | 1.7   | 1.1   | 1.6   | 1.8   |
|                                                                        | PrecStd     | 10.3        | 3.0   | 2.7   | 3.2   | 2.4   | 2.4   | 2.1   | 1.5   | 1.5   | 2.0   | 3.6   | 2.3   | 3.7   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 67.6  | 66.6  | 66.5  | 69.7  | 73.6  | 77.0  | 81.0  | 82.6  | 80.5  | 76.8  | 73.6  | 69.4  |
|                                                                        |             | MCWB        | 62.6  | 61.2  | 61.4  | 62.4  | 66.3  | 69.9  | 72.4  | 72.7  | 72.3  | 70.1  | 66.9  | 64.6  |
|                                                                        | 2%          | DB          | 66.3  | 65.2  | 65.1  | 67.4  | 71.0  | 74.8  | 79.2  | 80.9  | 78.5  | 75.0  | 71.5  | 67.9  |
|                                                                        |             | MCWB        | 62.2  | 60.7  | 60.7  | 61.6  | 65.0  | 68.5  | 71.7  | 72.5  | 71.7  | 69.4  | 66.1  | 63.4  |
|                                                                        | 5%          | DB          | 65.2  | 64.3  | 64.3  | 66.0  | 69.4  | 73.3  | 78.1  | 79.4  | 77.1  | 73.6  | 69.8  | 66.5  |
|                                                                        |             | MCWB        | 61.4  | 60.5  | 60.3  | 61.1  | 64.1  | 67.6  | 71.0  | 72.2  | 71.0  | 68.6  | 65.5  | 62.5  |
|                                                                        | 10%         | DB          | 64.3  | 63.0  | 63.0  | 64.6  | 67.9  | 71.7  | 76.8  | 78.5  | 75.8  | 72.3  | 68.4  | 65.7  |
|                                                                        |             | MCWB        | 60.6  | 59.5  | 59.4  | 60.2  | 63.0  | 66.7  | 70.2  | 71.8  | 70.3  | 67.9  | 64.5  | 61.9  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 63.9  | 63.8  | 63.6  | 64.3  | 67.5  | 71.5  | 74.3  | 75.1  | 74.5  | 71.9  | 68.8  | 65.9  |
|                                                                        |             | MCDB        | 66.1  | 64.9  | 64.9  | 66.9  | 70.7  | 74.7  | 78.5  | 79.2  | 77.6  | 74.7  | 71.2  | 68.2  |
|                                                                        | 2%          | WB          | 62.8  | 62.1  | 62.0  | 63.0  | 66.2  | 69.8  | 73.0  | 73.8  | 73.2  | 70.6  | 67.4  | 64.5  |
|                                                                        |             | MCDB        | 65.3  | 64.1  | 64.2  | 65.7  | 69.7  | 73.2  | 77.4  | 78.6  | 76.5  | 73.5  | 69.9  | 66.7  |
|                                                                        | 5%          | WB          | 62.0  | 61.2  | 61.0  | 62.0  | 65.0  | 68.7  | 72.0  | 73.2  | 72.3  | 69.7  | 66.3  | 63.4  |
|                                                                        |             | MCDB        | 64.5  | 63.4  | 63.3  | 64.8  | 68.5  | 72.1  | 76.4  | 78.1  | 75.7  | 72.8  | 69.0  | 65.9  |
|                                                                        | 10%         | WB          | 61.2  | 60.2  | 60.1  | 61.0  | 63.7  | 67.5  | 71.1  | 72.3  | 71.4  | 68.6  | 65.3  | 62.5  |
|                                                                        |             | MCDB        | 63.8  | 62.7  | 62.6  | 63.9  | 67.3  | 71.0  | 75.6  | 77.1  | 75.1  | 71.9  | 68.1  | 65.1  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 6.7   | 7.2   | 6.8   | 7.5   | 8.0   | 8.1   | 8.9   | 8.7   | 8.0   | 7.3   | 6.4   | 6.9   |
|                                                                        |             | MCDBR       | 7.1   | 7.5   | 7.4   | 8.9   | 9.8   | 9.1   | 9.8   | 9.9   | 8.8   | 7.8   | 7.3   | 7.4   |
|                                                                        | 5% WB       | MCWBR       | 6.7   | 7.0   | 6.6   | 7.0   | 6.8   | 6.5   | 6.2   | 6.2   | 6.1   | 6.1   | 6.1   | 7.2   |
|                                                                        |             | MCWBR       | 6.5   | 6.8   | 6.5   | 7.9   | 8.9   | 8.1   | 9.1   | 9.2   | 7.7   | 7.2   | 6.3   | 6.7   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.358 | 0.370 | 0.399 | 0.392 | 0.386 | 0.386 | 0.380 | 0.376 | 0.375 | 0.368 | 0.361 | 0.362 |
|                                                                        |             | taud        | 2.476 | 2.452 | 2.352 | 2.411 | 2.457 | 2.493 | 2.532 | 2.558 | 2.554 | 2.552 | 2.511 | 2.430 |
|                                                                        | Ebn at Noon |             | 255   | 268   | 271   | 281   | 283   | 283   | 282   | 276   | 266   | 252   | 242   | 242   |
|                                                                        |             | Edh at Noon | 26    | 30    | 36    | 36    | 35    | 34    | 32    | 31    | 29    | 27    | 24    | 25    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 602   | 866   | 1209  | 1613  | 1885  | 1993  | 2088  | 1942  | 1494  | 1037  | 681   | 524   |
|                                                                        | RadStd      |             | 53    | 49    | 84    | 139   | 132   | 154   | 116   | 89    | 68    | 59    | 38    | 31    |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page